SWAYAM DEVENDRA PANCHAL

Artificial Intelligence & Data Science Undergraduate | Mumbai, India

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SUMMARY

Third-year B.Tech student in Artificial Intelligence and Data Science (CGPA 9.70/10) who builds AI products end to end: LLM-assisted agents, retrieval pipelines, machine learning models and the full-stack web applications around them. 5x Hackathon Winner and 2x Ideathon Winner, and J.P. Morgan Chase Code for Good participant. Seeking a Data Science, ML, AI or Data Analyst internship.

TECHNICAL SKILLS

Languages and Databases: Python, SQL, JavaScript, SQLite, DBMS
Machine Learning and AI: Machine Learning, Deep Learning, NLP, Computer Vision, scikit-learn, TensorFlow, PyTorch, YOLOv9, Sentence Transformers, KeyBERT, Explainable AI (Grad-CAM), Gradient Boosting, Model Calibration
Generative AI and Agents: LLM integration (Claude API, Ollama), LangGraph multi-agent workflows, Retrieval-Augmented Generation (BM25 + embeddings), prompt and output guardrails, conversational copilots
Data Analytics: Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis, Statistical Analysis, Data Visualization, Dashboards, Power BI
Software and Tools: FastAPI, REST APIs, React.js, Chrome Extensions, GitHub Actions (CI), Docker, Git, GitHub, Vercel

PROFESSIONAL EXPERIENCE

AI Data Analyst | *InAmigos Foundation* | Jun 2026 - Jul 2026

- Performed data cleaning, preprocessing, exploratory data analysis and statistical analysis using Python, Pandas and NumPy to identify trends and patterns in organizational data.
- Built data visualizations, dashboards and analytical reports that communicated insights to non-technical stakeholders and supported data-driven decision-making.

PROJECTS

MIRAGE - Multimodal Scam and Deepfake Investigation Platform | *FastAPI, LangGraph, React, scikit-learn* | Team Outliers, D3COY 1.0 | github.com/SDP42/isacaps2

- Co-built a platform that judges whether a suspicious voice note, call, video, email or chat is safe to act on, fusing five detector modules (voice, face, provenance, text, identity) in a risk engine with policy rules and Shapley score explanations.
- Implemented a LangGraph multi-agent Tribunal (specialist, skeptic, judge, verifier and reporter agents) that explains every verdict with cited evidence IDs, backed by hybrid retrieval (BM25, TF-IDF and multilingual embeddings fused with Reciprocal Rank Fusion) over a casebook of past scams.
- Added optional Claude and local Llama layers with output guardrails that reject any rewrite that drops evidence tags or invents numbers, so the LLM explains results but never decides the risk score.
- Delivered a Chrome extension, a copilot with a trained question router (90.7% on 108 held-out questions), English/Hindi/Hinglish support and GitHub Actions CI.

Stratify - AI-Powered Algorithmic Trading Platform | *Python, React.js, SQL* | dombivli-hackathon-s6vb.vercel.app

- Developed a full-stack trading platform with market APIs for real-time ingestion, strategy execution and portfolio tracking, and a Pandas/NumPy backtesting framework (RSI, MACD, Moving Averages) to analyze risk-adjusted returns.

BrainMapAI - AI Knowledge Graph | *FastAPI, React.js, NetworkX* | github.com/SDP42/BrainMapAI

- Built a platform that turns unstructured PDFs and text into interactive, navigable knowledge graphs, addressing the context blindness of flat vector-search RAG by modeling concepts and relations as node-link structures that support multi-hop exploration.
- Engineered a modular NLP pipeline: layout-aware PDF extraction (PyMuPDF), text cleaning, hybrid concept extraction (spaCy syntax, KeyBERT, TF-IDF, RAKE, YAKE), acronym resolution and semantic deduplication with Sentence Transformers embeddings (all-MiniLM-L6-v2, cosine threshold 0.75).
- Implemented a rule-based relationship extractor using dependency parsing and subject-verb-object patterns with a relation taxonomy and confidence scoring, then built NetworkX graphs and computed density, degree and clustering statistics.
- Shipped a FastAPI backend (upload, process, graph and document APIs, SQLAlchemy and SQLite) and a React 18, TypeScript and ReactFlow canvas with zoom/pan exploration and insight views; backed by 215 automated tests, ADRs and benchmark write-ups.

Elevate 1.0 - National Hackathon Event Platform | *React.js, Vercel* | elevate1-0.vercel.app

- Co-built and deployed the official platform for a 24-hour national hackathon (2,500+ views, 700+ registrations) covering submissions, shortlisting, on-day check-in and meal-pass tracking for 115+ participants across 32 finalist teams, kept live throughout the event.

ACHIEVEMENTS AND LEADERSHIP

- **Hackathons:** 5x Hackathon Winner and 2x Ideathon Winner, building working AI, data and full-stack solutions under time constraints.
- **J.P. Morgan Chase and Co. Code for Good:** participant in the hackathon building technology solutions for non-profit organizations.
- **Leadership:** Secretary, DJS NSDC x Infomatrix (DJSCE); co-led Elevate 1.0, a 24-hour national hackathon, managing events, sponsorships and a 47-member organizing committee.
- **Mentorship:** mentor juniors in Python, NumPy, Pandas, Matplotlib, Seaborn and core machine learning algorithms through hands-on sessions and notebooks.

EDUCATION

B.Tech in Artificial Intelligence and Data Science | *Dwarkadas J. Sanghvi College of Engineering, Mumbai University* | 2024 - 2028 | CGPA 9.70

HSC, Maharashtra State Board | *Vidyaniketan Junior College of Commerce and Science* | 2024 | 87.67%

SSC, Maharashtra State Board | *Shree Siddhi Vinayagar English High School* | 2022 | 93%